



Addis Ababa Science and Technology University

Department of Electromechanical Engineering

TECHNICAL DESIGN & CALCULATION DOCUMENTATION

Industrial Grain and Seed Destoner Machine

Prepared for:

Design & Fabrication Team

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September 2025

Executive Summary

This technical documentation details the mechanical engineering analysis, dual-deck sieve sizing, pneumatic suction fan control, and dual vibro-motor drive system for an industrial grain and seed destoner machine. Developed for manufacturing by the design team at **Bejai Ethio Industrial & Engineering Solutions PLC**, the machine utilizes a two-tier screening process combined with negative-pressure air fluidization. The upper deck (6 mm × 6 mm aperture) removes large stones, while the lower deck (2.5 mm × 2.5 mm aperture) retains clean seeds (4 mm) and sifts out undersized dirt particles. Mathematical calculations cover fluidization suction velocity (1.5 m/s), suction fan sizing (1.08 m³/s, 2.2 kW motor), dual vibro-motor centrifugal force generation (6.40 kN total), and VFD-driven suction airflow control. The complete 3D CAD model, including assembly, exploded views and key components, is presented in Sections 2 and 4.

Keywords: Dual-Deck Destoner, Screen Sieve Aperture, Pneumatic Suction Control, Dual Vibro-Motors, VFD Regulation, Seed Processing.

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1 Company Overview & Operational Context

1.1 Bejai Ethio Industrial & Engineering Solutions PLC

Bejai Ethio Industrial & Engineering Solutions PLC is an Ethiopian engineering and fabrication enterprise specializing in industrial machinery design, agro-processing equipment manufacturing, and automation systems integration.

1.2 Project Scope for Design Team

This document outlines the design specifications, dual-sieve geometry, vibro-motor force balancing, and control sequences required for workshop assembly and testing of the dual-deck grain destoner (destoner real assembly.SLDASM).

2 Industrial Project Specifications

2.1 Project Title

Design, Modeling, and Engineering Analysis of an Industrial Dual-Deck Grain and Seed Destoner Machine

2.2 Problem Statement & Justification

Unprocessed Ethiopian grains contain both oversized impurities (large stones, soil clumps, straw) and undersized particles (sand, dust, broken seed fragments). Single-stage destoners often carry over large particles or lose small seeds. A dual-sieve configuration coupled with controlled suction air currents ensures complete particle separation across three distinct size fractions.

2.3 Design Objectives

1. Size a two-tier screen deck (6 mm $\times$ 6 mm upper; 2.5 mm $\times$ 2.5 mm lower) for seed isolation (4 mm mean diameter).
2. Calculate terminal fluidization velocities and pneumatic suction fan specs.
3. Analyze excitation force and dynamic power for twin counter-rotating vibro-motors.
4. Implement an integrated control logic for suction fan speed (VFD) and vibro-motor synchronization.

2.4 Machine Overview

Figure 1 shows the complete destoner assembly. Raw grain is stored in the elevated square feed hopper, which is supported on a welded box-section stand. The material is metered through the inclined feed pipe onto the upper deck of the dual-deck vibrating screen. The screen box is mounted on four isolation springs and is covered by a hood; the suction duct on top of the hood draws air upward through both decks.

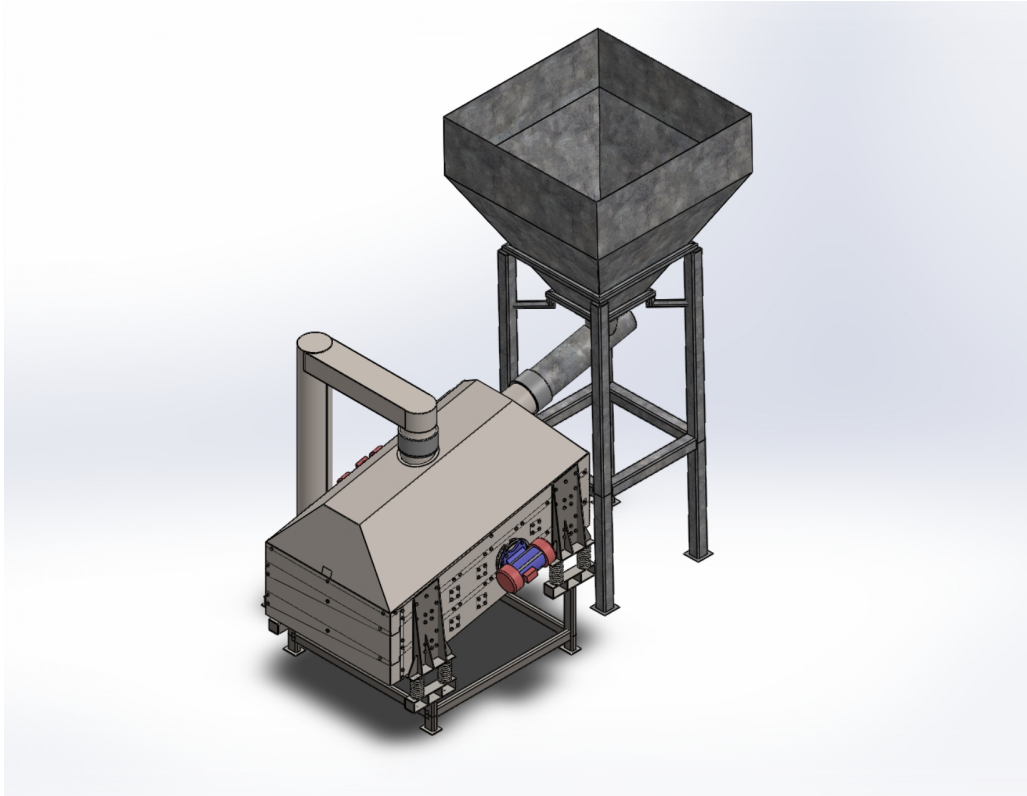


Figure 1: Complete destoner assembly: feed hopper with support stand, inclined feed pipe, dual-deck vibrating screen with suction hood and exhaust duct, twin vibro-motors, and spring-isolated base frame.

2.5 Engineering Calculations & Mechanical Design

2.5.1 Dual-Sieve Geometry & Separation Principle

The screening system consists of two inclined, stacked sieve decks:

- **Upper Deck Sieve (6 mm × 6 mm Square Mesh):** Designed aperture size $a_{top} = 6$ mm. Seed particles ($d_g = 4$ mm) and smaller impurities pass through the mesh onto the lower deck. Large stones, debris, and pods ($d > 6$ mm) are retained and discharged via the scalp discharge chute.
- **Lower Deck Sieve (2.5 mm × 2.5 mm Square Mesh):** Designed aperture size $a_{bottom} = 2.5$ mm. Clean seeds ($d_g = 4$ mm) are retained on top of this screen and conveyed upward by linear vibration to the clean seed outlet. Small sand particles, dust, and broken seeds ($d < 2.5$ mm) drop through into the bottom catch hopper.

Each screen panel is a framed square-mesh panel with a folded side flange drilled for bolt clamping to the deck side plates (Figure 2). The flange allows quick replacement of worn or blinded panels and provides the stiffness needed to keep the mesh tensioned under vibration.

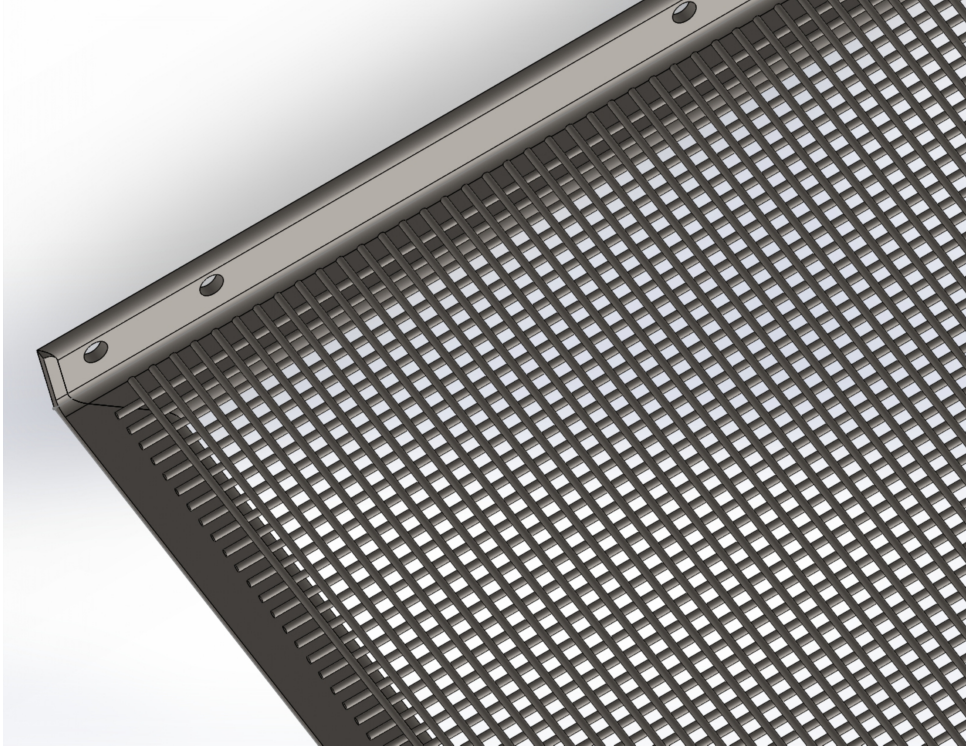


Figure 2: Close-up of the square-mesh sieve panel showing the woven square apertures, the folded edge flange, and the clamping bolt holes.

2.5.2 Fluidization Velocity & Air Suction Calculations

Air is drawn upward through both decks to loosen (fluidize) the seed bed so that denser particles such as stones stay in contact with the deck while the seeds float. The air velocity through the deck must therefore be *high enough to fluidize the seeds, low enough not to fluidize the stones, and far below the seed terminal velocity* so that seeds are not carried into the suction duct:

$$v_{mf,g} < v_a < v_{mf,s}, \quad v_a \ll v_{t,g} \quad (1)$$

where v_{mf} is the minimum fluidization velocity of grain (g) and stone (s), and v_t is the terminal velocity.

Design Parameters:

- Target seed diameter $d_g = 0.004$ m (4 mm)
- Seed bulk density $\rho_b = 750$ kg/m³ (used only for bed weight)
- Seed particle (true) density $\rho_{p,g} \approx 1300$ kg/m³
- Representative stone: $d_s = 0.005$ m, $\rho_s \approx 2600$ kg/m³
- Air: $\rho_a = 1.2$ kg/m³, $\mu = 1.81 \times 10^{-5}$ Pa s
- Gravity $g = 9.81$ m/s²; drag coefficient $C_d \approx 0.44$

Step 1: Seed terminal velocity (upper limit).

$$v_{t,g} = \sqrt{\frac{4 g d_g (\rho_{p,g} - \rho_a)}{3 C_d \rho_a}} = \sqrt{\frac{4 \times 9.81 \times 0.004 \times (1300 - 1.2)}{3 \times 0.44 \times 1.2}} \approx 11.3 \text{ m/s} \quad (2)$$

Check: $Re = \rho_a v_t d_g / \mu \approx 3000$, which is inside the range where $C_d \approx 0.44$ is valid.

Step 2: Minimum fluidization velocities (Wen–Yu correlation).

$$Ar = \frac{d^3 \rho_a (\rho_p - \rho_a) g}{\mu^2}, \quad Re_{mf} = \sqrt{33.7^2 + 0.0408 Ar} - 33.7, \quad v_{mf} = \frac{Re_{mf} \mu}{\rho_a d} \quad (3)$$

$$\text{Seed: } Ar = 2.99 \times 10^6, \quad Re_{mf} = 317 \Rightarrow v_{mf,g} \approx 1.20 \text{ m/s} \quad (4)$$

$$\text{Stone: } Ar = 1.17 \times 10^7, \quad Re_{mf} = 657 \Rightarrow v_{mf,s} \approx 1.98 \text{ m/s} \quad (5)$$

Step 3: Selected suction velocity and airflow.

$$v_{mf,g} = 1.20 < \mathbf{v_a = 1.50 \text{ m/s}} < v_{mf,s} = 1.98 \ll v_{t,g} = 11.3 \text{ m/s} \quad (6)$$

Total deck screen area $A_{deck} = 0.6 \text{ m} \times 1.2 \text{ m} = 0.72 \text{ m}^2$:

$$Q = A_{deck} \cdot v_a = 0.72 \text{ m}^2 \times 1.50 \text{ m/s} = 1.08 \text{ m}^3/\text{s} \text{ (3,888 m}^3/\text{h)} \quad (7)$$

Step 4: Required static pressure. The fluidized bed supports its own weight, so the bed pressure drop is $\Delta P_{bed} = \rho_b g H$. For a bed depth $H = 0.03 \text{ m}$:

$$\Delta P_{bed} = 750 \times 9.81 \times 0.03 \approx 221 \text{ Pa} \quad (8)$$

Component	ΔP (Pa)	Basis
Fluidized seed bed ($H = 30 \text{ mm}$)	221	Calculated
Two screen decks	50	Estimated
Hood, duct, elbow, entry/exit losses	250	Estimated
Sum	521	
With 1.25 margin	651	
Selected fan static pressure	700	Rounded up

Table 1: Suction system pressure budget.

Step 5: Fan power and motor selection.

$$P_{air} = Q \cdot \Delta P = 1.08 \times 700 = 756 \text{ W} \quad (9)$$

$$P_{motor} = \frac{P_{air}}{\eta_{fan} \eta_{belt}} = \frac{756}{0.65 \times 0.95} \approx 1.22 \text{ kW} \quad (10)$$

Applying a design safety factor of 1.25:

$$P_{motor, design} = 1.22 \times 1.25 \approx 1.53 \text{ kW} \Rightarrow \text{select a } \mathbf{2.2 \text{ kW, 3-phase induction motor}} \quad (11)$$

The suction is produced by a backward-curved centrifugal impeller housed in a scroll casing and driven by a three-phase induction motor through a V-belt drive (Figure 3). The belt drive lets the fan speed be matched to the required airflow by selecting the pulley ratio, while the VFD provides fine regulation during operation.

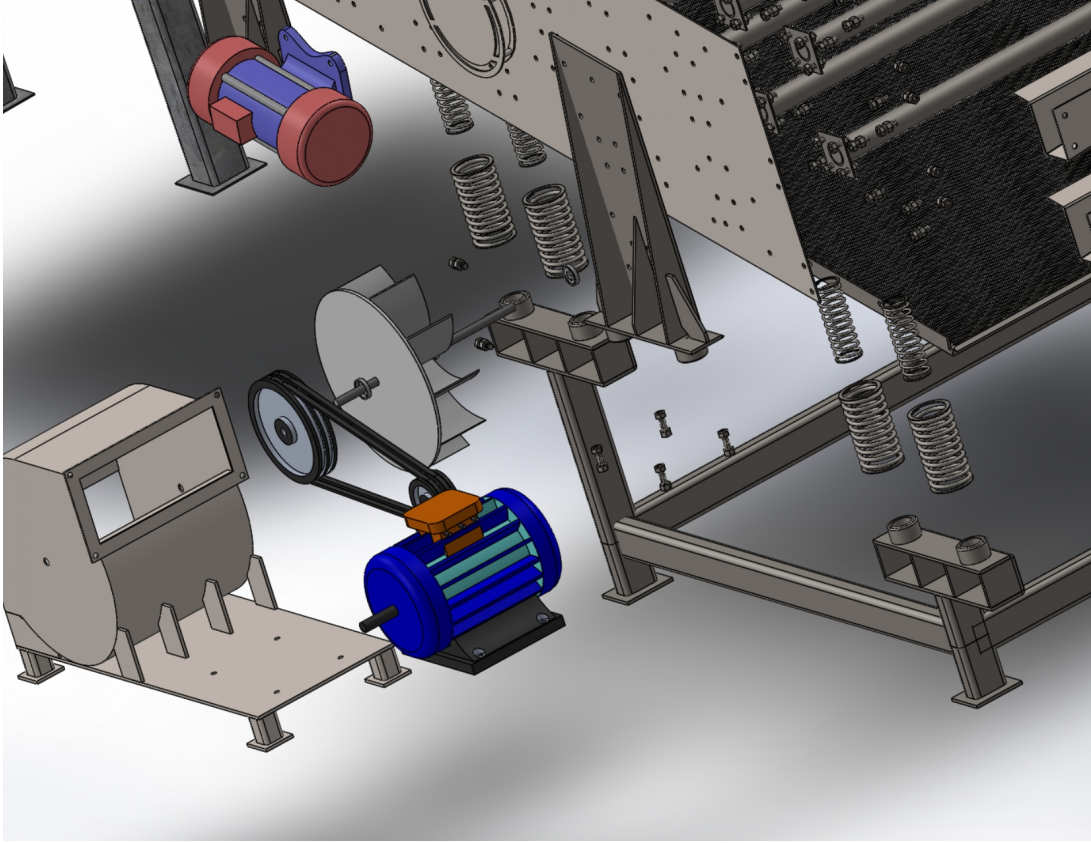


Figure 3: Exploded detail of the suction fan unit: backward-curved centrifugal impeller on its shaft, V-belt pulleys and belt, three-phase induction drive motor on its base, scroll fan casing on its mounting frame, together with the isolation springs, spring seats and one vibro-motor.

2.5.3 Dual Vibro-Motor Drive Dynamics

Linear directional vibration is generated by two identical counter-rotating unbalance vibro-motors mounted at a 45° excitation angle relative to the deck frame.

- Total Oscillating Mass (Decks + Frames + Screen Material): $m_{total} = 120 \text{ kg}$
- Operational Speed: $n = 900 \text{ RPM}$ ($\omega = 94.25 \text{ rad/s}$)
- Stroke Amplitude: $r = 0.006 \text{ m}$ (6 mm)

Peak dynamic acceleration a_{max} :

$$a_{max} = r \cdot \omega^2 = 0.006 \text{ m} \times (94.25 \text{ rad/s})^2 = 53.30 \text{ m/s}^2 \quad (12)$$

Total required centrifugal excitation force F_{dyn} :

$$F_{dyn} = m_{total} \cdot a_{max} = 120 \text{ kg} \times 53.30 \text{ m/s}^2 = 6396 \text{ N} \approx 6.40 \text{ kN} \quad (13)$$

Per-motor centrifugal force output F_{motor} :

$$F_{motor} = \frac{F_{dyn}}{2} = \frac{6.40 \text{ kN}}{2} = 3.20 \text{ kN per motor} \quad (14)$$

The two vibro-motors are bolted to heavy motor-mounting brackets on the deck side plates and drive the whole screen box, which rests on four coil isolation springs seated on the base frame (Figure 4). The spring isolation prevents the vibration from being transmitted to the floor and the supporting structure.

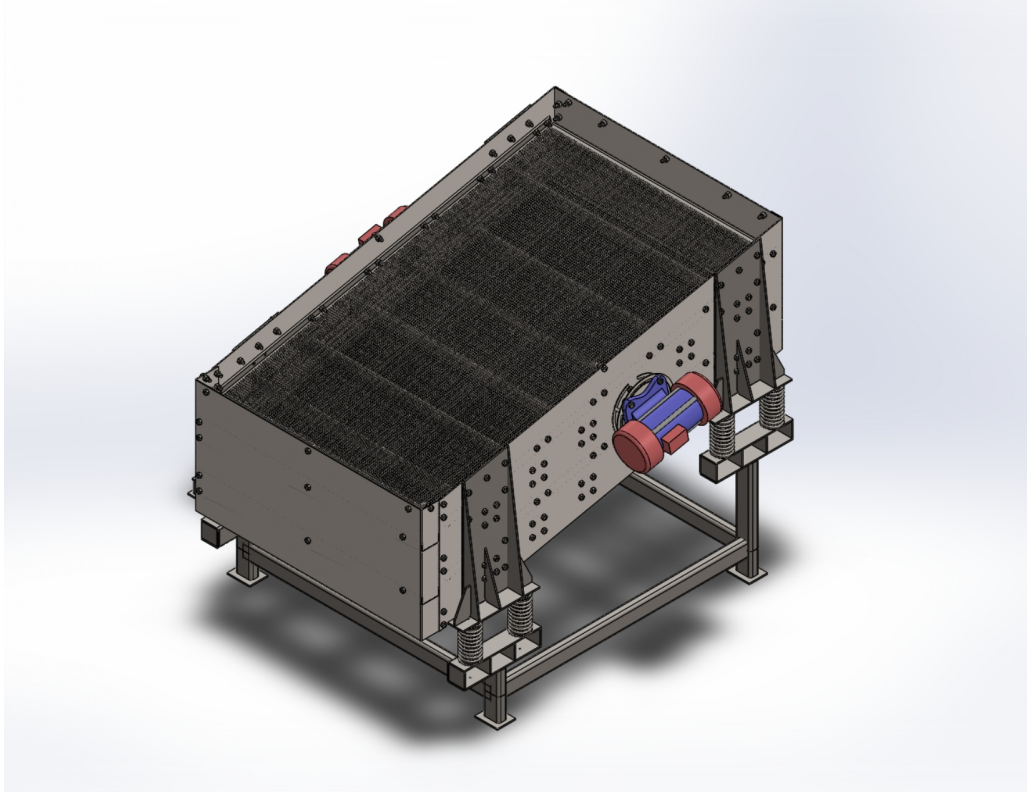


Figure 4: Vibrating screen unit (hood removed) showing the dual-deck screen box, side-mounted vibro-motor with its mounting bracket, gusseted spring supports, four coil isolation springs, and the welded base frame.

2.5.4 Power Calculations & Motor Procurement

Dynamic power requirements for twin vibro-motors with damping factor $\eta_{damp} = 0.12$:

$$P_{dyn} = \frac{1}{2} m_{total} \cdot r^2 \cdot \omega^3 \cdot \eta_{damp} = 0.217 \text{ kW} \quad (15)$$

Factoring in bed friction, inclination work, and mechanical load ($P_{load} = 1.25 \text{ kW}$):

$$P_{shaft} = P_{dyn} + P_{load} = 0.217 \text{ kW} + 1.25 \text{ kW} = 1.467 \text{ kW} \quad (16)$$

Applying a design safety factor S.F. = 1.5:

$$P_{total \text{ motor}} = 1.467 \text{ kW} \times 1.5 = 2.20 \text{ kW} \quad (17)$$

Hardware Specifications for Fabrication Team:

- **Vibro-Motors:** Two (2) 1.1 kW (1.5 HP each), 3-Phase, 6-Pole Vibro-Motors producing 3.20 kN unbalance force each at 900 RPM (2.2 kW combined capacity).
- **Suction Fan Assembly:** Backward-curved centrifugal exhaust fan rated for 1.08 m³/s (3,888 m³/h) at 700 Pa static pressure, driven by a 2.2 kW, 3-phase induction motor.

3 Electrical Automation & Motor Control System

3.1 Control System Architecture

The electrical panel controls three main actuators: the Air Suction Fan and the two counter-rotating Vibro-Motors.

Actuator	Device	Power Rating	/	Control Mechanism
Suction Air Fan	Motor	2.2 kW		VFD Frequency Speed Regulation (0–50 Hz)
Vibro-Motor 1	1.1 kW	Direct Line		Contactor w/ Overload Relays
Vibro-Motor 2	1.1 kW	Interlocked		Contactor (Synchronized)
Feed Hopper Gate	Solenoid / Pneumatic	On/Off		Limit Switch Control

Table 2: Actuators and control mechanisms.

3.2 Air Suction Fan Control Mechanism

- **VFD Speed Regulation:** The exhaust suction fan is driven via a Variable Frequency Drive (VFD). Operators adjust the airflow velocity (0–1.5 m/s) on the control panel based on seed density and grain moisture content.
- **Air Pressure Feedback:** A differential pressure transmitter (ΔP) installed inside the suction hood continuously measures suction pressure to prevent bed choking or over-suction.

3.3 Twin Vibro-Motor Control Mechanism

- **Counter-Rotating Synchronization:** The two vibro-motors are wired to operate in opposite rotational directions. Self-synchronization causes transverse forces to cancel out, providing a pure linear oscillatory trajectory along the deck.
- **Thermal Protection & Phase Monitoring:** Thermal overload relays and a phase loss monitor ensure both vibro-motors start and shut down simultaneously. If one vibro-motor trips, the control system instantly shuts down the second motor to prevent catastrophic torsional damage to the deck frame.

3.4 Sequential System Interlocking Logic

1. **Start Command** → VFD ramps up the Air Suction Fan to target speed (1.5 m/s velocity).
2. **Pressure Verification** → Differential pressure switch confirms stable negative pressure → 3-second safety delay.
3. **Vibro-Motors Start** → Twin vibro-motors engage simultaneously in counter-rotation.
4. **Material Feed Open** → Hopper solenoid gate opens; uncleaned seed mix enters the upper 6 mm deck.
5. **Emergency Fault Routine** → Motor overload, imbalance, or loss of suction pressure instantly closes the feed gate and trips all drives.

4 Fabrication Output & CAD Verification

The primary assembly is saved as `destoner_real_assembly.SLDASM`. Workshop fabrication drawings specify frame reinforcements, isolation spring mounts, and double-deck screen clamping mechanisms. The figures below show the CAD model used for verification of fits, clearances and assembly sequence.

4.1 Complete Assembly

See Figure 1 for the full machine layout (hopper, feed pipe, screen unit, hood and suction duct).

4.2 Exploded Views

Figure 5 shows the complete exploded assembly: the feed hopper and stand with its feed pipe, the dual-deck screen box with both screens lifted out, the vibro-motors, the isolation springs and spring seats, the base frame, and the suction hood, duct and fan housing. Figure 6 focuses on the screen box: the upper screen panel is lifted off to expose the tubular cross-members that support the decks, the lower screen panel, the deck side plates with bolt patterns, the gusseted spring brackets, and the circular vibro-motor mounting opening in the side plate.

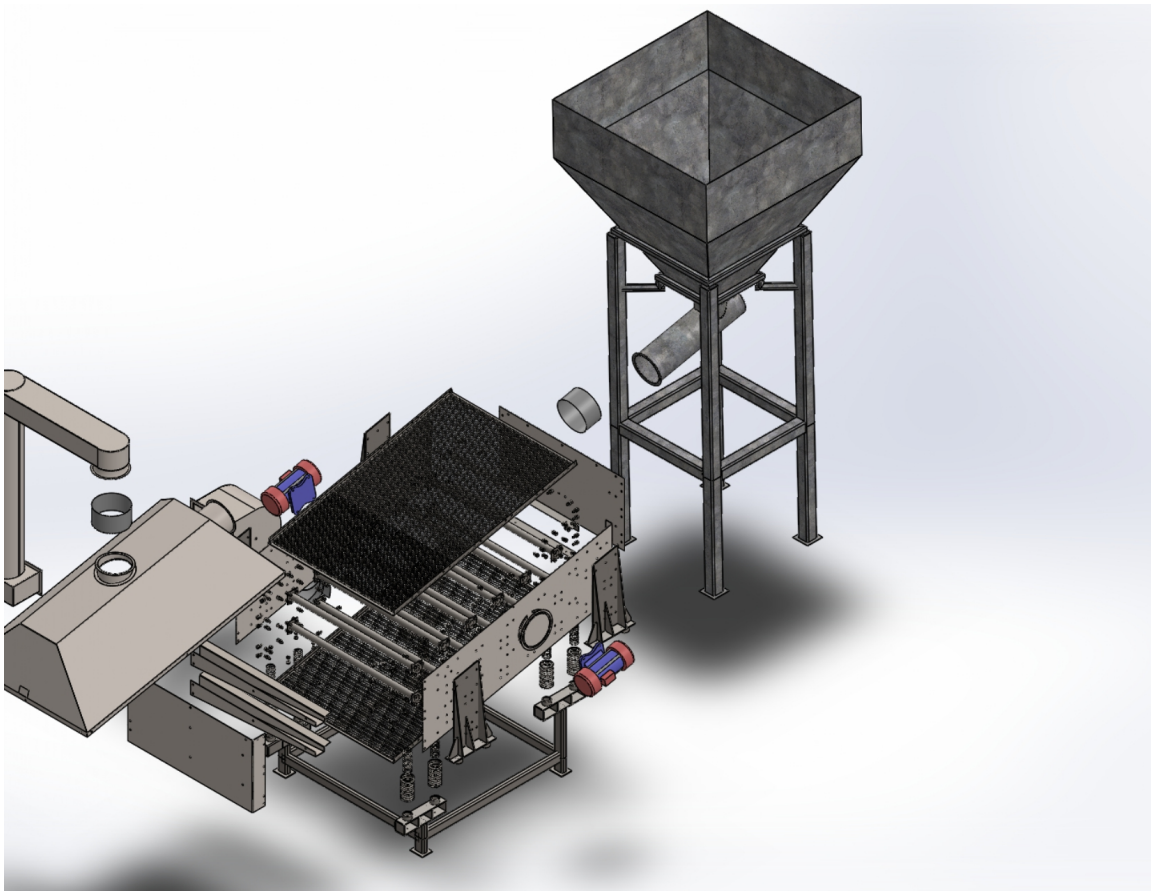


Figure 5: Full exploded view of the dual-deck destoner assembly.

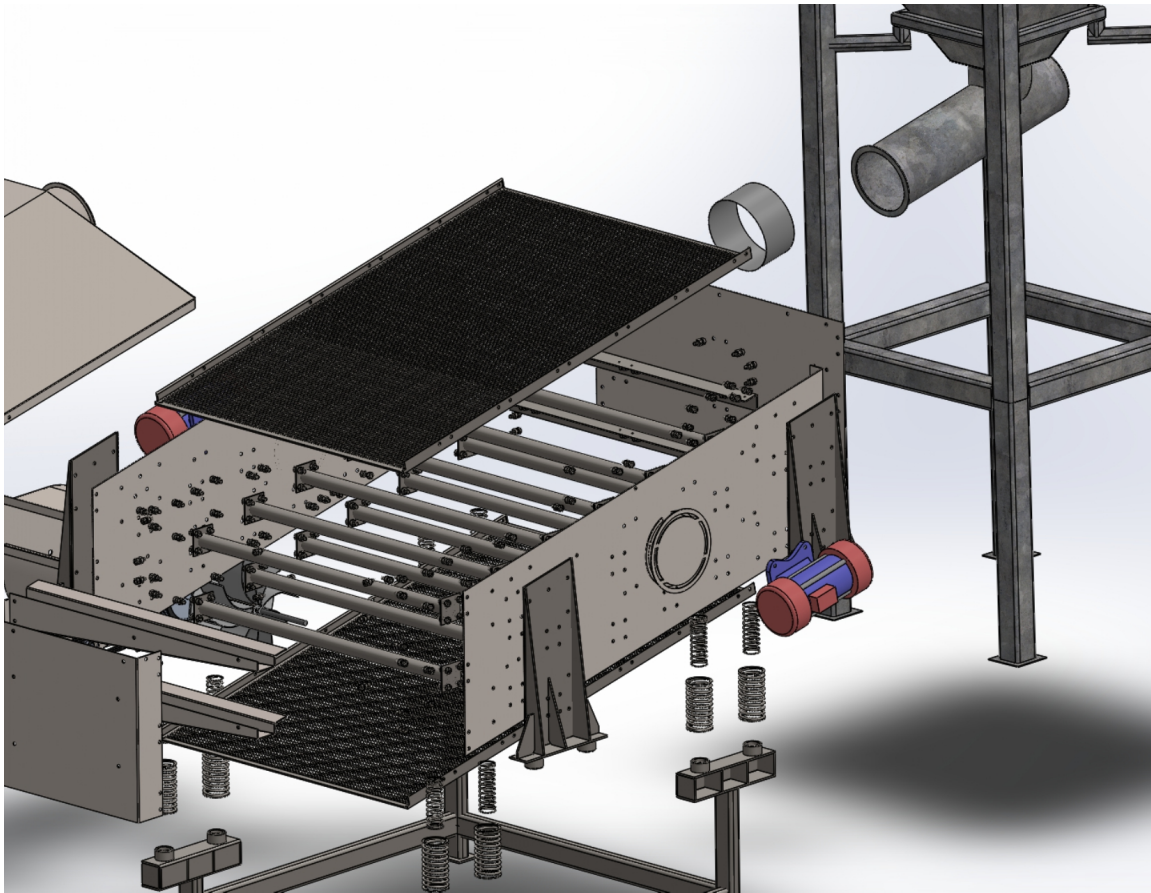


Figure 6: Exploded view of the vibrating screen box showing the upper screen panel, lower screen panel, tubular deck cross-members, side plates, vibro-motor mounting, isolation springs and spring seats on the base frame.

4.3 Key Components

The key sub-assemblies are shown in the figures introduced earlier:

- Sieve panels — Figure 2
- Vibrating screen unit and vibro-motor — Figure 4
- Suction fan impeller, belt drive and induction motor — Figure 3